

Algebra 2
REVIEW Unit 8

Name: KBY Period:

Find for which values of x is each expression undefined.

1. $\frac{x+7}{7x-28}$
 $\hookrightarrow 7(x-4)=0$
 $\boxed{x=4}$

2. $\frac{x^2+7x-18}{x^2-100}$
 $\hookrightarrow (x+10)(x-10)=0$
 $\boxed{x=-10, 10}$

Find the indicated feature of each function. If there is none, state "none."

3. The x -intercept:

$f(x) = \frac{9x-5}{x+8}$
 $9x-5=0$
 $9x=5$
 $x=\frac{5}{9}$
 $\boxed{(\frac{5}{9}, 0)}$

4. The y -intercept:

$f(x) = \frac{6x^2-11x-3}{x^2-2x-4}$
 $f(0) = \frac{-3}{-4} = \frac{3}{4}$
 OR
 $\boxed{(0, \frac{3}{4})}$

5. The vertical asymptote(s):

$f(x) = \frac{x^2-81}{x^2+3x-18}$
 $= \frac{(x+9)(x-9)}{(x+6)(x-3)}$
 $\Rightarrow \boxed{\text{VA @ } x=-6 \text{ and } x=3}$

6. The horizontal asymptote:

$f(x) = \frac{x^3-1}{x^2+5x-6}$
 "Top Heavy"
 $\boxed{\text{none}}$

7. The coordinates of the hole:

$f(x) = \frac{x^2+6x+5}{x^2+8x+7}$
 $= \frac{(x+5)(x+1)}{(x+7)(x+1)}$
 Hole @ $x=-1$
 $y = \frac{(-1+5)}{(-1+7)} = \frac{4}{6} = \frac{2}{3}$
 $\boxed{(-1, \frac{2}{3})}$

8. The horizontal asymptote:

$f(x) = \frac{3-2x+6x^2}{2x^2+1}$ OUT OF ORDER!
 $\frac{6x^2}{2x^2}$ "EQUAL"
 $\boxed{y=3}$

State any horizontal or vertical asymptotes, holes, intercepts, or say "none." Then put them on the graph and try to sketch the graph.

9. $y = \frac{3x^2-3}{x^2+5x-6}$

$$= \frac{3(x^2-1)}{(x+6)(x-1)} = \frac{3(x+1)(x-1)}{(x+6)(x-1)} = \frac{3(x+1)}{x+6} = \frac{3x+3}{x+6}$$

Hole @ $x=1$
 $y = \frac{3(1+1)}{1+6} = \frac{6}{7}$ $(1, \frac{6}{7})$

a) Holes:

$(1, \frac{6}{7})$

b) Vertical Asymptotes:

$x = -6$

c) Horizontal Asymptotes:

$y = \frac{3}{1}, y = 3$

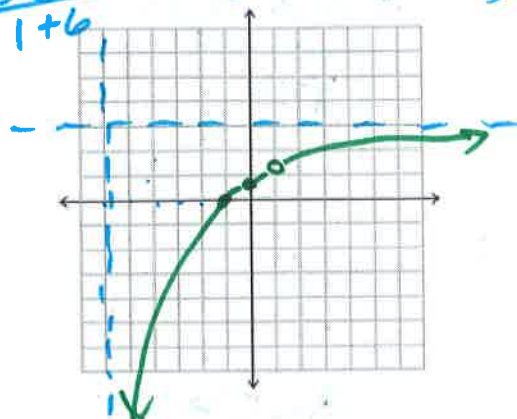
d) x-intercepts:

$(-1, 0)$

e) y-intercepts:

$(0, \frac{1}{2})$

f) Sketch the graph, including all of its features.



10.

$$y = \frac{x+2}{x^2+3x+2} = \frac{(x+2)}{(x+2)(x+1)} = \frac{1}{x+1}$$

Hole @ $x = -2$
 $y = \frac{1}{-2+1} = \frac{1}{-1} = -1$

a) Holes:

$(-2, -1)$

b) Vertical Asymptotes:

$x = -1$

c) Horizontal Asymptotes:

$y = 0$ "bottom heavy"

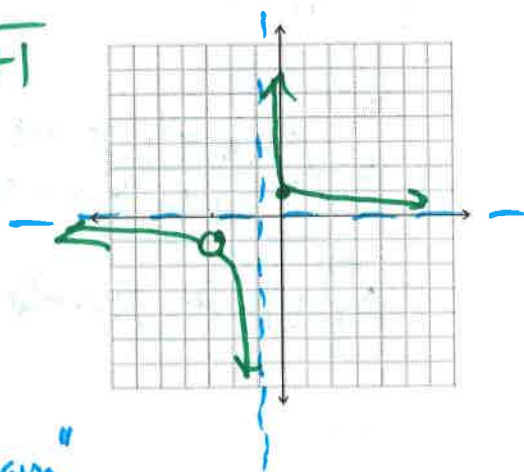
d) x-intercepts:

none

e) y-intercepts:

$(0, 1)$

f) Sketch the graph, including all of its features.



Simplify the rational expressions.

$$11. \frac{2x^2-72}{x^2-5x-6} = \frac{2(x^2-36)}{(x-6)(x+1)}$$

$$= \frac{2(x+6)(x-6)}{(x-6)(x+1)}$$

$$= \boxed{\frac{2(x+6)}{(x+1)}}$$

$$12. \frac{16x^3}{y^3} \cdot \frac{xy^2}{8x^2} = \frac{2x^4y^2}{x^2y^3}$$

$$= \boxed{\frac{2x^2}{y}}$$

$$13. \frac{x+5}{x^2+2x-24} \cdot \frac{x^2-36}{x^2+10x+25}$$

$$= \frac{(x+5) \cdot (x+6)(x-6)}{(x+6)(x-4)(x+5)(x+5)}$$

$$= \boxed{\frac{(x-6)}{(x-4)(x+5)}}$$

$$14. \frac{5x+10}{x^2-16} \div \frac{x^2+8x+12}{x-3}$$

$$= \frac{5(x+2)}{(x+4)(x-4)} \cdot \frac{(x-3)}{(x+6)(x+2)}$$

$$= \boxed{\frac{5(x-3)}{(x+4)(x-4)(x+6)}}$$

$$15. \frac{3x+1}{x+3} + \frac{2x-1}{2x+6}$$

$$= \frac{2(3x+1)}{2(x+3)} + \frac{2x-1}{2(x+3)}$$

$$= \frac{6x+2+2x-1}{2(x+3)}$$

$$= \boxed{\frac{8x+1}{2(x+3)}}$$

$$16. \frac{5x+2}{x-7} + \frac{3}{x}$$

$$= \frac{x(5x+2)}{x(x-7)} + \frac{3(x-7)}{x(x-7)}$$

$$= \frac{5x^2+2x+3x-21}{x(x-7)}$$

$$= \boxed{\frac{5x^2+5x-21}{x(x-7)}}$$

$$17. \frac{4}{x^2-16} - \frac{5}{x^2+2x-24}$$

$$= \frac{(x+6)4}{(x+6)(x+4)(x-4)} - \frac{5}{(x+6)(x-4)(x+4)}$$

$$= \frac{4x+24 - (5x+20)}{(x+6)(x+4)(x-4)}$$

$$= \frac{-x+4}{(x+6)(x+4)(x-4)} = \frac{-1(x-4)}{(x-4)(x+4)(x+6)} = \boxed{\frac{-1}{(x+4)(x+6)}}$$

$$18. \frac{3}{x+2} - 5$$

$$= \frac{3}{x+2} - \frac{5(x+2)}{1(x+2)}$$

$$= \frac{3-5x-10}{(x+2)} = \boxed{\frac{-5x-7}{x+2}}$$